Case Study 5: Investigating DJI Mavic Air Flight Evidence

1. Case reference and student identification

Field	Entry
Full name	Abubakar Mustapha Sadiq
Registration number	2025/FWSD/11202
Course code	SBT-DF204 Computer Forensics Case Studies
Case title	Investigating DJI Mavic Air Flight Evidence (Case Study 5)
Examination	3 November 2026, Kali Linux virtual machine, offline, using the course artefacts only
Supporting files	Tool output text files: https://github.com/Bigmusty01/SBT-DF204-Investigating-DJI-Mavic-Air-Flight-Evidence (folder evidence-log)

2. Executive conclusion

I was able to examine one of the four sources: the Android logical image, Android_Logical.zip. It turned out to be an incomplete download, but the part I could read holds a consistent set of DJI GO 4 records from 19 June 2018. These are a main flight record, a cached video with its metadata, two detailed flight records and a few cached pictures. The phone's own metadata puts its clock in EDT (UTC-4). On that basis the cached video started at 18:50:39 UTC and ran for about four minutes, the main flight record started at 18:50:34 UTC, and the first detailed record began at about 18:47:38 UTC and was last written at about 18:58 UTC.

My confidence is **moderate** that DJI GO 4 on this phone logged one flight session in that window, because the file names, the sidecar metadata and the file times all agree with one another. My confidence is **low** about the route, the exact take-off and landing, and any link to the microSD card. I could not decode the TXT and DAT files, and the raw microSD and EWF images were not part of what I received. What the records show is an application and a phone clock. They do not show who was holding the controller, and I do not name or attribute anyone.

3. Evidence acquisition and integrity

The only evidence file available to me was Android_Logical.zip, found in the OneDrive folder Image (Figure S04). No mobile forensic report, raw microSD image or EWF image came with it, so those three sources are listed below as not received.

Source	Original name	Bytes	SHA-256	Source location	Working copy
Android logical image	Android_Logical.zip	413,261,824	49629ec3259fad25e5dccd51f72cfa9d88f4f253d0051447043ee4591d635862	OneDrive, folder Image; browser download saved 2026-11-03 00:12	~/case5/working/Android_Logical.z extracted in ~/case5/working/android/
Mobile forensic report	not received	n/a	n/a	n/a	n/a
Raw internal microSD image	not received	n/a	n/a	n/a	n/a
EWF image	not received	n/a	n/a	n/a	n/a

I moved the download into a folder called originals and made it read-only with chmod 444. Before doing anything else I recorded its size and SHA-256 (Figure S01). I then made a working copy with cp -p and confirmed its hash was identical to the original (Figure S03). Every extraction and every look inside the archive was done on the working copy. The tools were the standard Kali Linux packages: sha256sum, 7z, unzip, exiftool, ffprobe, strings, xxd and file. I did not write down their version strings while I was working, so I do not quote them here.

```
┌──(kali㉿kali)-[~/case5/working]
└─$ mv ~/Downloads/Android_Logical.zip ~/case5/originals/
cd ~/case5/originals
chmod 444 Android_Logical.zip
ls -l --time-style=full-iso | tee ../out/01_sizes.txt
sha256sum Android_Logical.zip | tee ../out/01_sha256_before.txt
total 403580
-r--r--r-- 1 kali kali 413261824 2026-11-03 00:12:11.610060697 -0500 Android_Logical.zip
49629ec3259fad25e5dccd51f72cfa9d88f4f253d0051447043ee4591d635862  Android_Logical.zip

┌──(kali㉿kali)-[~/case5/originals]
└─$ unzip -l Android_Logical.zip | tee ../out/01_zip_listing.txt | head -60
  End-of-central-directory signature not found.  Either this file is not
  a zipfile, or it constitutes one disk of a multi-part archive.  In the
  latter case the central directory and zipfile comment will be found on
  the last disk(s) of this archive.
Archive:  Android_Logical.zip
unzip:  cannot find zipfile directory in one of Android_Logical.zip or
        Android_Logical.zip.zip, and cannot find Android_Logical.zip.ZIP, period.

┌──(kali㉿kali)-[~/case5/originals]
└─$
```

Figure S01. *The original moved into originals/ and set read-only; size 413,261,824 bytes and SHA-256 recorded. unzip -l then fails on the archive.*

The archive is incomplete

unzip could not open the file and reported that the end-of-central-directory signature was missing (Figure S01). 7z l could still list the entries but ended with “Unexpected end of archive” and a physical size of 413261824 (Figure S02). When I extracted the working copy, 7z recovered 373 files and then stopped with a data error at Android_Logical/sdcard/Samsung/Music/Over the Horizon.mp3 (Figure S03). Everything stored after that entry in the archive is missing.

OneDrive lists the file as 396 MB (Figure S04), while my copy is 413,261,824 bytes, about 394 MiB. That points to an incomplete download, although the OneDrive figure may be rounded. I did not have a complete replacement copy during this work. All the DJI GO 4 artefacts discussed below were inside the readable part. Files that sit after the cut-off could not be examined, and I do not treat their absence as evidence that they never existed.

```
Listing archive: Android_Logical.zip

--
Path = Android_Logical.zip
Type = zip
ERRORS:
Unexpected end of archive
Physical Size = 413261824
Characteristics = Local

   Date      Time    Attr         Size   Compressed  Name
------------------- ----- ------------ ------------  ------------------------
2018-10-11 15:14:24 D....            0            0  Android_Logical
2018-10-03 11:57:16 D....            0            0  Android_Logical/property
1980-01-01 00:00:00 .....           24           24  Android_Logical/property/net.hostname
1980-01-01 00:00:00 .....           16           16  Android_Logical/property/ro.serialno
2018-10-03 11:58:06 D....            0            0  Android_Logical/sdcard
2018-09-22 22:30:28 D....            0            0  Android_Logical/sdcard/Alarms
2018-10-03 11:57:16 D....            0            0  Android_Logical/sdcard/amap
2018-10-03 11:57:18 D....            0            0  Android_Logical/sdcard/amap/openamaplocationsdk
2017-10-10 09:56:42 .....           20           20  Android_Logical/sdcard/amap/openamaplocationsdk/a.dat
2017-10-10 09:59:38 .....        24576         1321  Android_Logical/sdcard/amap/openamaplocationsdk/alsn.db
2017-10-09 12:20:28 .....        12824          346  Android_Logical/sdcard/amap/openamaplocationsdk/alsn.db-journal
2018-06-19 11:13:42 .....        20480         1168  Android_Logical/sdcard/amap/openamaplocationsdk/alsn20170807.db
2018-06-19 11:12:42 .....        12824          315  Android_Logical/sdcard/amap/openamaplocationsdk/alsn20170807.db-journal
2018-10-03 11:57:28 D....            0            0  Android_Logical/sdcard/Android
2018-10-03 11:57:24 D....            0            0  Android_Logical/sdcard/Android/data
2018-10-03 11:57:18 D....            0            0  Android_Logical/sdcard/Android/data/com.amazon.dee.app
2018-09-22 22:30:34 D....            0            0  Android_Logical/sdcard/Android/data/com.amazon.dee.app/cache
2018-10-03 11:57:18 D....            0            0  Android_Logical/sdcard/Android/data/com.android.browser.provider
2018-09-22 22:30:34 D....            0            0  Android_Logical/sdcard/Android/data/com.android.browser.provider/files
2018-10-03 11:57:18 D....            0            0  Android_Logical/sdcard/Android/data/com.android.calendar
2018-09-22 22:30:34 D....            0            0  Android_Logical/sdcard/Android/data/com.android.calendar/files
[Android_Logical.zip]
  End-of-central-directory signature not found.  Either this file is not
  a zipfile, or it constitutes one disk of a multi-part archive.  In the
  latter case the central directory and zipfile comment will be found on
  the last disk(s) of this archive.
unzip:  cannot find zipfile directory in one of Android_Logical.zip or
        Android_Logical.zip.zip, and cannot find Android_Logical.zip.ZIP, period.

┌──(kali㉿kali)-[~/case5/originals]
└─$
```

Figure S02. *7z l lists the archive but reports “Unexpected end of archive” with physical size 413261824.*



Figure S03. *The working copy has the same SHA-256 as the original. Extraction of the working copy stops with a data error at Samsung/Music/Over the Horizon.mp3.*

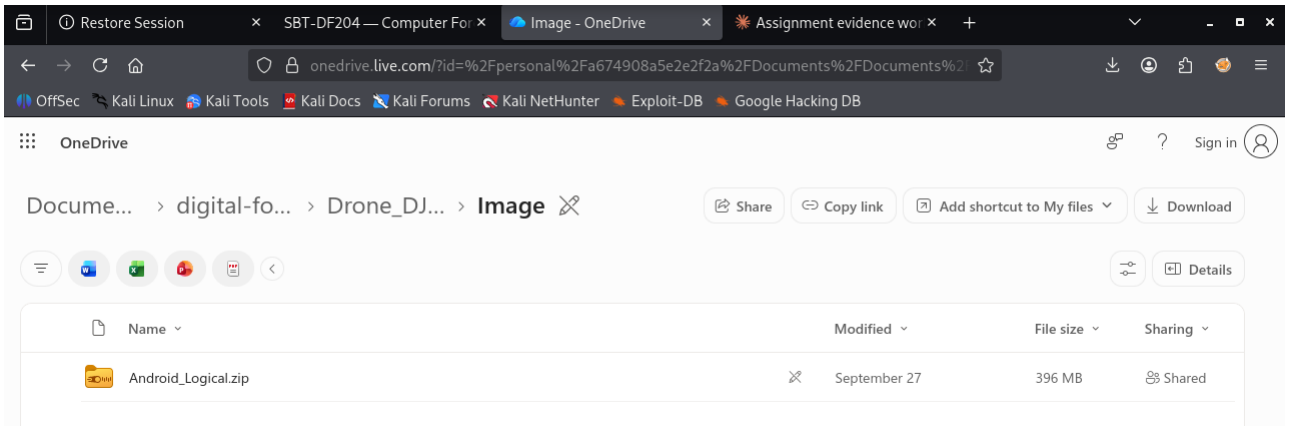


Figure S04. *Source of the file: OneDrive folder Image containing Android_Logical.zip, listed at 396 MB.*

4. Method

Finding the artefacts. I searched the extracted tree for paths containing dji.go.v4 and then looked at each family of file in turn. Before opening anything I recorded its size, its timestamps and its SHA-256. The cached images, the video and the text files were identified with file, exiftool, ffprobe, strings and xxd, and the image cache journal was read as plain text. I did not run or open any artefact in a way that could change the working copy, and nothing was uploaded to a drone portal or a flight parser.

Time. There are three clocks in this evidence and I kept them apart. The first is the time written into file names by the app on the phone. The second is the modification time stored in the zip. A zip stores no time zone: `7z l -slt` printed the same value under TZ=UTC and under TZ=America/New_York (Figure S15). The third is the time on the extracted files, which the extraction tool converted with a fixed offset. For the main flight record the stored time 12:55:40 became 17:55:40 under TZ=UTC, so I did not treat extracted file times as UTC.

The phone's time zone comes from the sidecar .info file of the video. Its first line is #Tue Jun 19 14:55:41 EDT 2018, a header written by the phone, and its CaptureDate of 14:50:39 matches the video file name (Figure S10). I therefore read file-name times as phone-local EDT, so UTC is the file-name time plus four hours. The same header shows that the stored zip times run two hours behind the phone clock: the sidecar's stored time is 12:55:42 against a header

time of 14:55:41. I used that two-hour gap for the stored times, so UTC for a stored time is the stored time plus six hours.

Raw microSD, EWF and comparison. I did not receive these images, so I did not do this part of the work. I did not infer anything about the microSD card from the phone data.

5. Findings

Q1. How the case materials were preserved (E01)

Section 3 and Figures S01 to S04 cover this. The Android image was preserved read-only with matching original and working-copy hashes, and its truncation is documented. I could not preserve or record the other three sources because I did not receive them.

Q2. DJI GO 4 artefacts present (E02)

The DJI GO 4 data is in three places under Android_Logical/sdcard/: the DJI/dji.go.v4 folder on shared storage (flight records, video cache and logs), the app's own Android/data/dji.go.v4 folder (image cache) and Android/obb/dji.go.v4 (an expansion file). Figures S05 and S06 show the search, and Figure S07 shows sizes, times and hashes for the main artefacts.

Family	Path (under Android_Logical/sdcard/)	Bytes	SHA-256
Main flight record (TXT)	DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt	555,776	0a9f1816d52a98f840f6e67c08a14473ae5a0a2e75e9484d5de66368c4fd4
Detailed record FLY057 (DAT)	DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT	4,002,791	06130820a4a813bf21f7286150bd3c83084105fecc899bb4d8534895fa2df
Detailed record FLY058 (DAT)	DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-59-05_FLY058.DAT	1,104,150	fdfccf0c5b1d90d189c19032bb34e151914f42fba42c7994d0492cf327ade0
Cached video (MP4)	DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4	57,791,765	2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab
Video sidecar	DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.info	1,441	a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b
Video analytics (MAP)	DJI/dji.go.v4/DJI_RECORD/analytics/2018_06_19_14_50_39.map	143,358	db58496a49d6f8ae82af46338cfcd397f5b7b741b1f726718a69caf5d466d0
Video analytics (TLV)	DJI/dji.go.v4/DJI_RECORD/analytics/2018_06_19_14_50_39.tlv	444,914	6cf36ce052645c6a4df9ecca10499347919048cefa503f0631a1d6b9dc007
Cached picture (WebP)	Android/data/dji.go.v4/cache/uil-images/2cmsgpo3xyz7a5g8c2twm1zcj0	8,414	084c66e1b40f1728dc242b1ff63b854fbdfd9c9aef15ad7648c5d401f7610e
Cached picture (JPEG 1920×550)	Android/data/dji.go.v4/cache/uil-images/2ec8zc1rfdpe2wf7ownjl19c10	666,994	e8c47a7bbebc7e292b4ad1579356c374f5418ad6a586ce235b681fe5a9bfc
Cached picture (JPEG 994×332)	Android/data/dji.go.v4/cache/uil-images/6bkm680zskvd7si9m75pippui0	19,605	850c4e407a0dffc60c33714b0389bdc0bdf7a4197920f5e7d1c35f74bc84c
Cache journal	Android/data/dji.go.v4/cache/uil-images/journal	523	b6fd7c963b1f48f86fb128d6ababc8ed9d49d136bcfee90c14e64276cebcfa

Several other media files turned up in the search, and I left them out of the analysis on purpose. The visual_guide and videoguide folders hold DJI tutorial videos, and editor/music/cover holds image assets for the app's video editor. All of them are stamped 12:07:32 to 12:07:34 on 2018-06-19, which looks like the app unpacking its own resources (Figure S08). There is also an older recording, DJI/dji.pilot/DJI_RECORD/2017_10_10_10_02_47.mp4 (207,448,934 bytes, SHA-256 ce078b575399110fb022fac5c4802bf50dfe0f58e76976480c93aefee354e5dd, Figure S13), from October 2017. It belongs to a different day, so I excluded it. The databases folder under dji.go.v4 is empty.

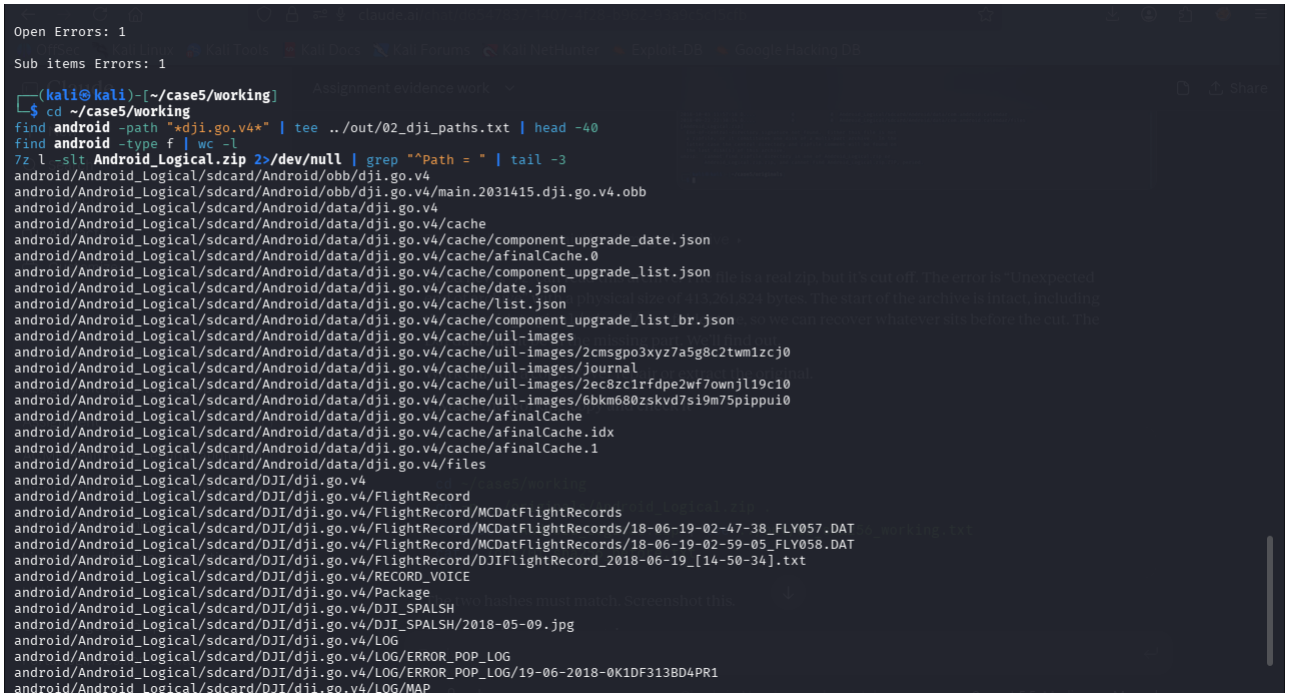


Figure S05. *Search of the extracted tree for dji.go.v4 paths: the start of the listing, including the obb file and the cache folders.*

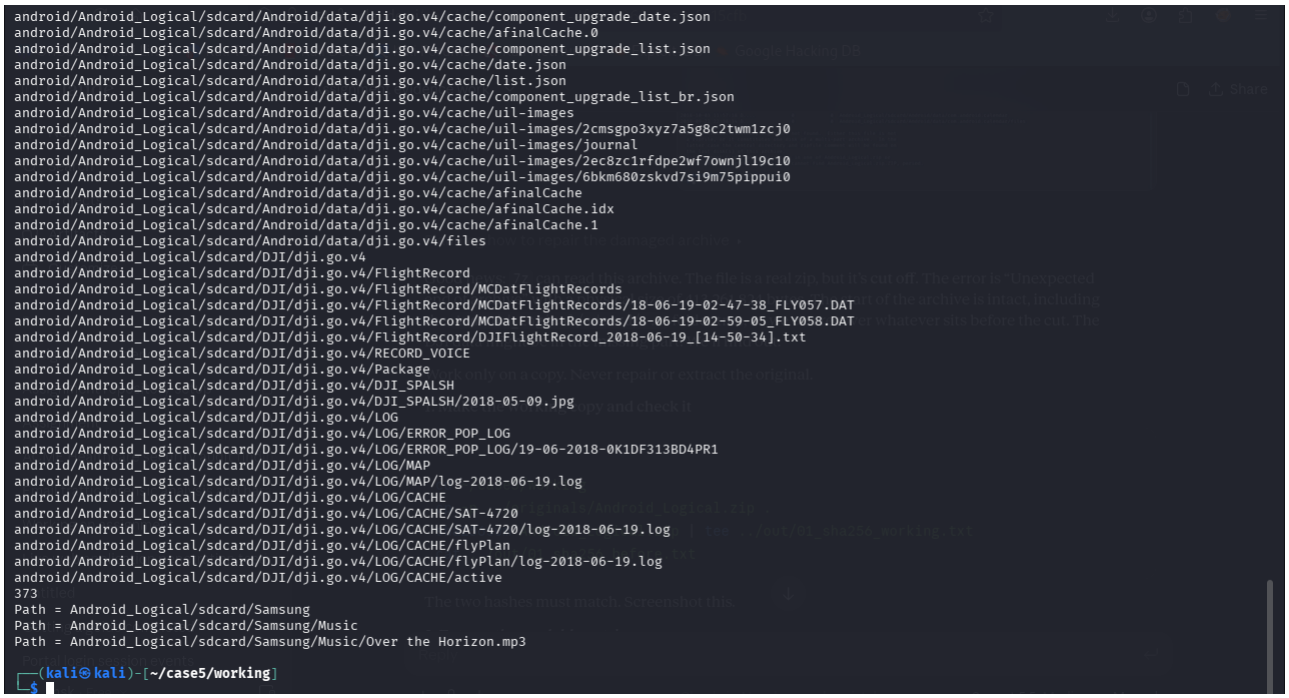


Figure S06. *End of the same listing: FlightRecord, MCDatFlightRecords and LOG entries. 373 files were extracted in total.*

```
android/Android_Logical/sdcard/DJI/dji.go.v4/LOG/CACHE/active
373
Path = Android_Logical/sdcard/Samsung
Path = Android_Logical/sdcard/Samsung/Music
Path = Android_Logical/sdcard/Samsung/Music/Over the Horizon.mp3

┌──(kali㉿kali)-[~/case5/working]
└─$ D=android/Android_Logical/sdcard
find $D/DJI/dji.go.v4/FlightRecord $D/Android/data/dji.go.v4/cache/uil-images $D/DJI/dji.go.v4/DJI_SPLASH -type f \
  -exec ls -l --time-style=full-iso {} \; | tee ../out/02_dji_files.txt
find $D/DJI/dji.go.v4/FlightRecord $D/Android/data/dji.go.v4/cache/uil-images $D/DJI/dji.go.v4/DJI_SPLASH -type f \
  -exec sha256sum {} \; | tee ../out/02_dji_hashes.txt
-rw-rw-r-- 1 kali kali 4002791 2018-06-19 13:58:24.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02
-47-38_FLY057.DAT
-rw-rw-r-- 1 kali kali 1104150 2018-06-19 14:02:12.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02
-59-05_FLY058.DAT
-rw-rw-r-- 1 kali kali 555776 2018-06-19 13:55:40.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-
50-34].txt
-rw-rw-r-- 1 kali kali 8414 2018-06-19 12:12:38.000000000 -0400 android/Android_Logical/sdcard/Android/data/dji.go.v4/cache/uil-images/2cmsgpo3xyz7a5g8c2tw
m1zcj0
-rw-rw-r-- 1 kali kali 523 2018-06-19 12:12:42.000000000 -0400 android/Android_Logical/sdcard/Android/data/dji.go.v4/cache/uil-images/journal
-rw-rw-r-- 1 kali kali 666994 2018-06-19 12:12:42.000000000 -0400 android/Android_Logical/sdcard/Android/data/dji.go.v4/cache/uil-images/2ec8zc1rfdpe2wf7ow
njl19c10
-rw-rw-r-- 1 kali kali 19605 2018-06-19 12:12:40.000000000 -0400 android/Android_Logical/sdcard/Android/data/dji.go.v4/cache/uil-images/6bkm680zskvd7si9m75
pippui0
find: 'android/Android_Logical/sdcard/DJI/dji.go.v4/DJI_SPLASH': No such file or directory
06130820a4a813bf21f7286150bd3c83084105fecc899bb4d8534895fa2df3a1  android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-
47-38_FLY057.DAT
fdfccf0c5b1d90d189c19032bb34e151914f42fba42c7994d0492cf327ade07a  android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-
59-05_FLY058.DAT
0a9f1816d52a98f840f6e67c08a14473ae5a0a2e75e9484d5de66368c4fd4c4a  android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-
50-34].txt
084c66e1b40f1728dc242b1ff63b854fbdfd9c9aef15ad7648c5d401f7610e2c  android/Android_Logical/sdcard/Android/data/dji.go.v4/cache/uil-images/2cmsgpo3xyz7a5g8c2
twm1zcj0
b6fd7c963b1f48f86fb128d6ababc8ed9d49d136bcfee90c14e64276cebcfa67  android/Android_Logical/sdcard/Android/data/dji.go.v4/cache/uil-images/journal
e8c47a7bbebc7e292b4ad1579356c374f5418ad6a586ce235b681fe5a9bfc694  android/Android_Logical/sdcard/Android/data/dji.go.v4/cache/uil-images/2ec8zc1rfdpe2wf7ow
njl19c10
850c4e407a0dffc60c33714b0389bdc0bdf7a4197920f5e7d1c35f74bc84c79e  android/Android_Logical/sdcard/Android/data/dji.go.v4/cache/uil-images/6bkm680zskvd7si9m7
5pippui0
find: 'android/Android_Logical/sdcard/DJI/dji.go.v4/DJI_SPLASH': No such file or directory

┌──(kali㉿kali)-[~/case5/working]
└─$
```

Figure S07. *Sizes, extracted times and SHA-256 of the main flight record, the two DAT files and the cached picture files.*

```
-rw-rw-r-- 1 kali kali 668050 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/SPRING_FESTIVAL.png
-rw-rw-r-- 1 kali kali 144504 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/CHEERS.png
-rw-rw-r-- 1 kali kali 156390 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/FOREST_TRAILS.png
-rw-rw-r-- 1 kali kali 25065 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/HAPPY_BOSSA_NOVA.png
-rw-rw-r-- 1 kali kali 149800 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/CANT_WAIT.png
-rw-rw-r-- 1 kali kali 141053 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/RUSH.png
-rw-rw-r-- 1 kali kali 105583 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/IN_THE_SKY.png
-rw-rw-r-- 1 kali kali 27942 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/THE_BRIGHTEST_LIGHT.png
-rw-rw-r-- 1 kali kali 187454 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/CALL_OF_THE_FOREST.png
-rw-rw-r-- 1 kali kali 62872 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/GOLDEN_SKIES.png
-rw-rw-r-- 1 kali kali 159184 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/BREAK_IT.png
-rw-rw-r-- 1 kali kali 33033 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/INITIATION.png
-rw-rw-r-- 1 kali kali 130265 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/DRIVING.png
-rw-rw-r-- 1 kali kali 149136 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/BEYOND_THE_HORIZON.png
-rw-rw-r-- 1 kali kali 175487 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/STARLINE.png
-rw-rw-r-- 1 kali kali 491736 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/HAPPY_NEW_YEAR.png
-rw-rw-r-- 1 kali kali 119100 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/HIGHWAY_SHUTTLE.png
-rw-rw-r-- 1 kali kali 115492 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/SUMMER_PARTY.png
-rw-rw-r-- 1 kali kali 42421 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/IBIZA_NIGHTS.png
-rw-rw-r-- 1 kali kali 153097 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/SINKING_DEEPS.png
-rw-rw-r-- 1 kali kali 98162 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/RUNNING.png
-rw-rw-r-- 1 kali kali 129962 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/WALK_ABOUT.png
-rw-rw-r-- 1 kali kali 35808 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/LEGION.png
-rw-rw-r-- 1 kali kali 112474 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/IMMEASURABLE.png
-rw-rw-r-- 1 kali kali 149722 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/SO_GOOD.png
-rw-rw-r-- 1 kali kali 102730 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/FOLLOWBEATS.png
-rw-rw-r-- 1 kali kali 42154 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/FALLING_DEEP.png
-rw-rw-r-- 1 kali kali 164622 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/BRIGHT_CITY_LIGHTS.png
-rw-rw-r-- 1 kali kali 143349 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/LOVE_THE_DAY.png
-rw-rw-r-- 1 kali kali 813952 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/tapfly_free.mp4
-rw-rw-r-- 1 kali kali 526161 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/parallel_tracking_tutorial.mp4
-rw-rw-r-- 1 kali kali 965299 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/tapfly_coordinate.mp4
-rw-rw-r-- 1 kali kali 534579 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/spotlight_tutorial.mp4
-rw-rw-r-- 1 kali kali 582669 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/quick_shot_planet_tutorial.mp4
-rw-rw-r-- 1 kali kali 640358 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/tapfly_backward.mp4
-rw-rw-r-- 1 kali kali 562070 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/quick_shot_circle_tutorial.mp4
-rw-rw-r-- 1 kali kali 628589 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/quick_shot_rocky_tutorial.mp4
-rw-rw-r-- 1 kali kali 542202 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/tapfly_forward.mp4
-rw-rw-r-- 1 kali kali 557418 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/quick_shot_diagonal_tutorial.mp
4
-rw-rw-r-- 1 kali kali 513666 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/trace_tutorial.mp4
-rw-rw-r-- 1 kali kali 501297 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/quick_shot_spiral_tutorial.mp4
-rw-rw-r-- 1 kali kali 693274 2018-06-19 12:07:32.000000000 -0400 android/Android_Logical/sdcard/DJI/dji.go.v4/visual_guide/quick_shot_comet_tutorial.mp4
```

Figure S08. *Search for media files: the tutorial videos and editor image assets, all stamped 12:07:32 on 2018-06-19, which I excluded.*

Q3. What the main flight record establishes (E03)

The main flight record is DJIFlightRecord_2018-06-19_[14-50-34].txt, 555,776 bytes. file reports it only as “data”, and its first bytes are binary, so it cannot be read as text (Figure S09). No offline DJI log parser was available in the environment I was working in, so I did not decode it. That means I cannot report the aircraft or device identifiers, the motor-on, take-off or landing events, the GPS positions or the recorded duration.

What I can say comes from its name and its stored time. The name gives a start of 14:50:34 phone-local, which is **18:50:34 UTC**. Its stored last-write time of 12:55:40 plus six hours is **18:55:40 UTC**. That suggests a record of roughly five minutes, but this is an estimate from file times, not from the record’s content.

The time-zone anomaly is the two-hour gap between the zip's stored times and the phone's EDT clock, described in section 4. It holds for all five files I compared, but I worked it out from one clear data point, the sidecar header, so I would want the time fields inside the TXT and DAT to confirm it (Figures S14 and S15).

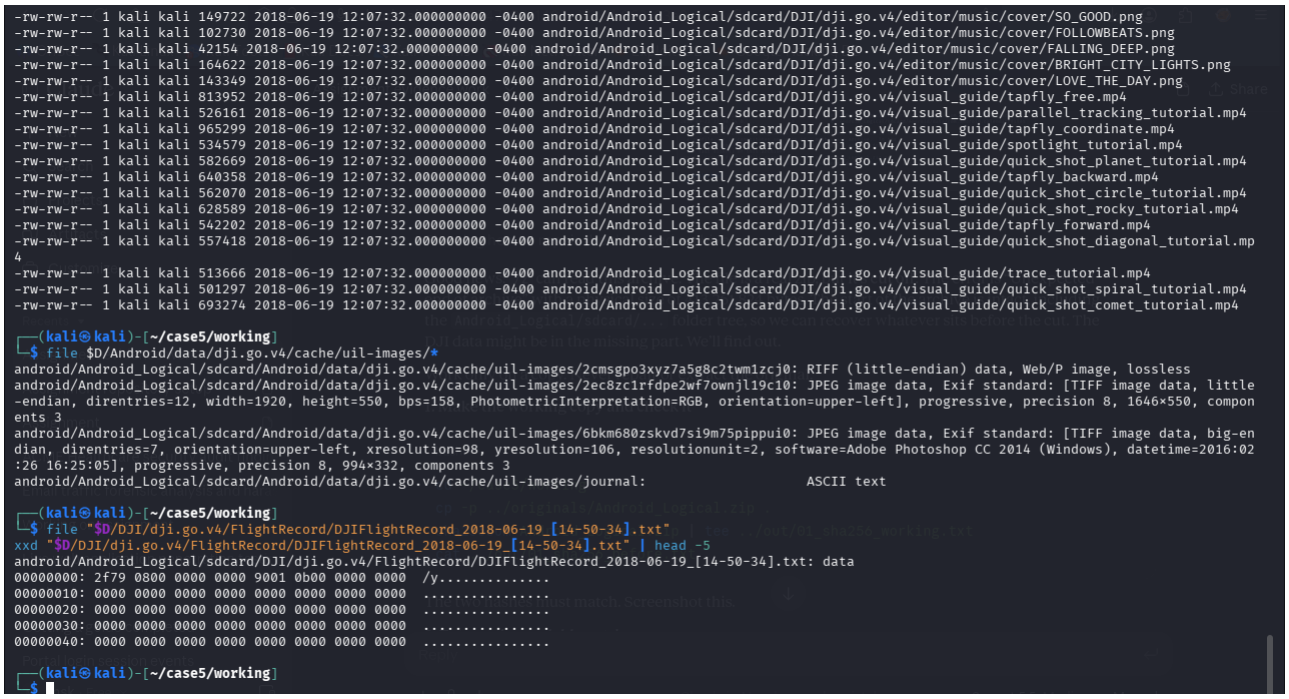


Figure S09. *file on the cached images (a WebP, two JPEGs and the journal as ASCII text), and the first bytes of the main flight record, which is binary.*

Q4. What the cached video establishes (E04)

The video is DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4, 57,791,765 bytes. I hashed it and read its metadata before looking at any content. It is H.264 (avc1) at 1280×720, 29.236 frames per second, with an average bitrate of 1.92 Mbps and a duration of 0:04:01 (241 seconds). The encoder tag is Lavf56.15.102. The track and media create and modify dates inside the file are all 0000:00:00 00:00:00 (Figure S11), so the video carries no absolute time of its own. Its time has to come from its file name and from the sidecar.

The sidecar .info fills that gap. It gives CaptureDate=2018/06/19 14:50:39, the source path /storage/emulated/0/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4, StartTimeMsec=0 and EndTimeMsec=234833. It also contains frame sizes (1920 wide on the drone side, 1280×720 locally), Is_HD=false, FrameJumped=14 and position fields, whose values I have redacted in Figure S10. Taking the start as 14:50:39 phone-local, the video began at **18:50:39 UTC**. It ended at 18:54:40 UTC going by the container duration, or 18:54:33.8 UTC going by EndTimeMsec. The two durations differ by about six seconds and I could not explain why.

Next to it in DJI_RECORD/analytics are a .map and a .tlv file with the same name prefix (Figure S12). I cannot say what they contain, so I only note that they share the video's name and its stored time. The video, the sidecar and both analytics files all have a stored last-write time of 12:55:42, within two seconds of the main flight record at 12:55:40. That suggests the app closed them together at the end of the session.

As for its relationship to the main flight record, the video starts five seconds after the record's file-name start and finishes before the record's last write, so it sits inside the record's time span. A 720p copy at under 2 Mbps with 1920-wide drone-side dimensions looks like a downscaled cached copy rather than the original recording. I could not confirm that without the microSD card.

```
drwxrwxr-x 2 kali kali   4096 2018-10-03 12:57:36.000000000 -0400 .
drwxrwxr-x 3 kali kali   4096 2018-10-03 12:57:36.000000000 -0400 ..
-rw-rw-r-- 1 kali kali 143358 2018-06-19 13:55:42.000000000 -0400 2018_06_19_14_50_39.map
-rw-rw-r-- 1 kali kali 444914 2018-06-19 13:55:42.000000000 -0400 2018_06_19_14_50_39.tlv
2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00  android/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a  android/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.info
db58496a49d6f8ae82af46338cfcd397f5b7b741b1f726718a69caf5d466dd6a  android/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/analytics/2018_06_19_14_50_39.map
6cf36ce052645c6a4df9ecca10499347919048cefa503f0631a1d6b9dc00792c  android/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/analytics/2018_06_19_14_50_39.tlv
#Tue Jun 19 14:55:41 EDT 2018
CameraType=23
Source_File_Path=/storage/emulated/0/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
FPS_Drone=30
EndTimeMsec=234833
LocalFileName=2018_06_19_14_50_39
StartTimeMsec=0
FolderID_Drone=100
WhiteBalance=0
ImageDescription=
Is_HD=false
ISO=0
PixelXDimension_Drone=1920
PixelYDimension_Local=720
Video_Type=0
ExposureMode=1
PixelXDimension_Local=1280
Sync_Drone_Time=2798,21406,31080,40573,47491,51419,56131,62052,66403,72863,82480,89239,95506,100138,102690,107245,110007,116680,120557,125491,132277,140057
,150696,157268,161168,169594,178082,184778,191034,197045,198421,200956,202643,204492,215218,219213,221189,226924,235297,241797,249261,260181,267023,271101,
275135,281403,287748,295237
Version=1.0
Video_Resolution_Enum_Drone=10
FPS_local=30
CaptureDate=2018/06/19 14\:50\:39
UUID_Drone=3890597672
File_Source_Type=1
FileID_Drone=1
ProductType=38
LocationString=
PositionRelativeAlt=12.0
PositionGPSLat: [redacted]
FrameJumpped=14

┌──(kali㉿kali)-[~/case5/working]
└─$
```

Figure S10. *SHA-256 of the 2018 video, its sidecar and the analytics files, followed by the sidecar content. The position value is redacted.*

```
[QuickTime]     MatrixStructure                 : 1 0 0 0 1 0 0 0 1
[QuickTime]     PreviewTime                     : 0 s
[QuickTime]     PreviewDuration                 : 0 s
[QuickTime]     PosterTime                      : 0 s
[QuickTime]     SelectionTime                   : 0 s
[QuickTime]     SelectionDuration               : 0 s
[QuickTime]     CurrentTime                     : 0 s
[QuickTime]     NextTrackID                     : 2
[QuickTime]     HandlerType                     : Metadata
[QuickTime]     HandlerVendorID                 : Apple
[Track1]        TrackHeaderVersion              : 0
[Track1]        TrackCreateDate                 : 0000:00:00 00:00:00
[Track1]        TrackModifyDate                 : 0000:00:00 00:00:00
[Track1]        TrackID                         : 1
[Track1]        TrackDuration                   : 0:04:01
[Track1]        TrackLayer                      : 0
[Track1]        TrackVolume                     : 0.00%
[Track1]        MatrixStructure                 : 1 0 0 0 1 0 0 0 1
[Track1]        ImageWidth                      : 1280
[Track1]        ImageHeight                     : 720
[Track1]        MediaHeaderVersion              : 0
[Track1]        MediaCreateDate                 : 0000:00:00 00:00:00
[Track1]        MediaModifyDate                 : 0000:00:00 00:00:00
[Track1]        MediaTimeScale                  : 90000
[Track1]        MediaDuration                   : 0:04:01
[Track1]        MediaLanguageCode               : und
[Track1]        HandlerType                     : Video Track
[Track1]        HandlerDescription              : VideoHandler
[Track1]        GraphicsMode                    : srcCopy
[Track1]        OpColor                         : 0 0 0
[Track1]        CompressorID                    : avc1
[Track1]        SourceImageWidth                : 1280
[Track1]        SourceImageHeight               : 720
[Track1]        XResolution                     : 72
[Track1]        YResolution                     : 72
[Track1]        BitDepth                        : 24
[Track1]        VideoFrameRate                  : 29.236
[ItemList]      Encoder                         : Lavf56.15.102
[Composite]     ImageSize                       : 1280×720
[Composite]     Megapixels                      : 0.922
[Composite]     AvgBitrate                      : 1.92 Mbps
[Composite]     Rotation                        : 0
```

Figure S11. *ExifTool output for the video: 1280×720, 0:04:01, zero create and modify dates, encoder Lavf56.15.102.*



Figure S12. *Zip entries for the analytics files of the 2018 recording: sizes, stored times, CRC values and offsets inside the archive.*

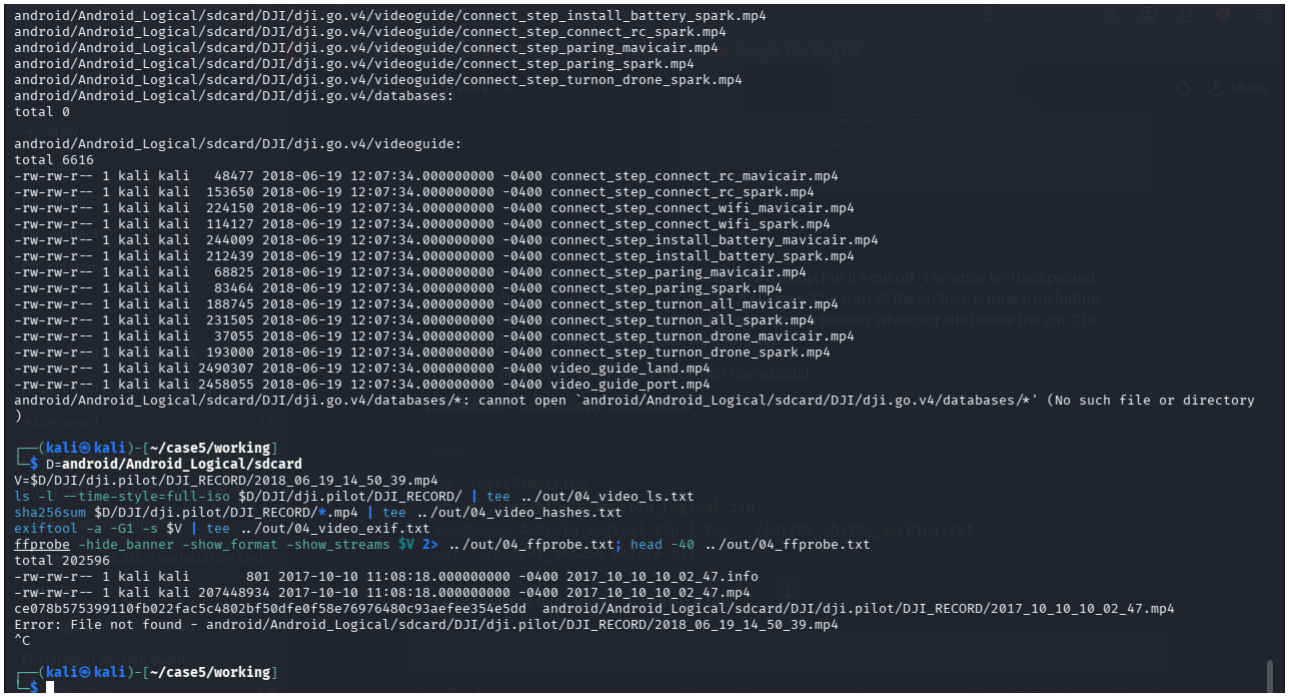


Figure S13. *The 2017 recording and its sidecar under DJI/dji.pilot/DJI_RECORD, with its SHA-256. The "File not found" lines at the bottom come from a mistyped folder in that command and are not a finding.*

Q5. What the detailed flight records establish (E05)

There are two detailed records, 18-06-19-02-47-38_FLY057.DAT (4,002,791 bytes) and 18-06-19-02-59-05_FLY058.DAT (1,104,150 bytes). As with the TXT, I could not decode them, so I cannot report power-on, motor-on, take-off, landing or power-off events from their content. I can only use their names and stored times.

The names carry no AM or PM marker. I read 02-47-38 as 14:47:38, which is 18:47:38 UTC. FLY057's stored last write of 12:58:24 becomes 18:58:24 UTC, so the record would span about 10 minutes 46 seconds. FLY058 starts at 14:59:05 (18:59:05 UTC) and was last written at 19:02:12 UTC, a span of about 3 minutes 7 seconds. The implied data rates are close to each other, about 6.2 KB/s for FLY057 and 5.9 KB/s for FLY058. If 02-47-38 meant 02:47 in the morning, FLY057 would have to be about twelve hours long, which does not fit a file of 4 MB. That supports the afternoon reading.

On this reading, the main flight record (18:50:34 to 18:55:40 UTC) and the video (18:50:39 to about 18:54:40 UTC) both fall inside FLY057, which is what I would expect from a record that runs from power-on to power-off. FLY058 begins about 41 seconds after FLY057's last write and looks like a separate session. I found no matching main record or video for it. All of this rests on file names and file times, so I treat it as provisional (Figures S07 and S14).

Q6. What the cached pictures establish (E06)

The folder Android/data/dji.go.v4/cache/uil-images holds three cached images and a journal file. They are a lossless WebP, a 1920×550 JPEG and a 994×332 JPEG. ExifTool shows that the two JPEGs were made in Adobe Photoshop (CC 2017 for the first, CC 2014 for the second) and carry creation dates of 2018-05-17 19:57:18 and 2016-02-26 16:25:05, both with a +08:00 offset. I saw no GPS or camera tags in the output (Figure S16).

The journal is a standard image-cache journal (libcore.io.DiskLruCache). Its entries give sizes of 8414, 19605 and 666994 bytes, which match the three files. The 1920-wide image was written, removed and written again, and the file name is the journal key followed by a 0. Going by the file times, the order of capture was 2cmsgpo… at 12:12:38, 6bkm680… at 12:12:40 and 2ec8zc… at 12:12:42 on the stored clock.

These look like banner-style images that the app downloaded and cached, not photos taken by a drone camera. They tell me the app was running and online at about that time. They cannot support any inference about a flight, a place or an operator, and the stored time should not be read as UTC.

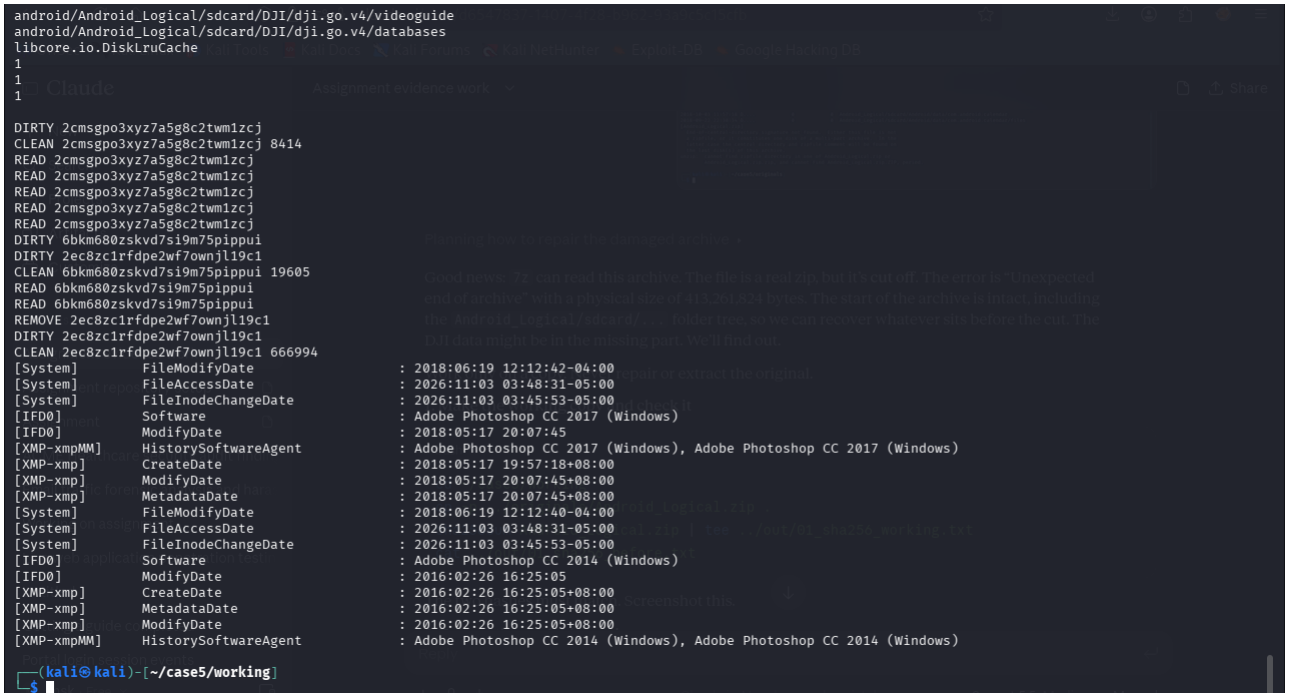


Figure S16. *The image-cache journal and ExifTool output for the two JPEGs: Photoshop software, 2016 and 2018 creation dates, no GPS tags.*

Q7. What the raw internal microSD image shows (E07)

I did not receive a raw microSD image, so I have no layout, file listing, hashes, timestamps or media identifier to report. I have left this question unanswered rather than guess from the phone data.

Q8. What the EWF image shows and how it compares (E08)

I did not receive an EWF image either, so I could not verify an image, record a serial number or compare hashes with a raw image. No comparison between the microSD sources and the mobile findings was possible.

Q9. The conclusion I can defend

Sections 2, 6 and 7 give the conclusion. In short, I can defend application-level activity on the phone and the timing of its records, and nothing about people or routes.

6. Timeline and correlation (UTC)

Every time below is provisional. Each comes either from a file-name time on the phone's EDT clock plus four hours, or from a stored last-write time plus six hours. They should be confirmed against the decoded TXT and DAT records.

Stage	UTC timestamp	ID	Source artefact	Observed fact	Interpretation and limit
Device state and power on	2018-06-19 18:47:38	E05	FLY057 DAT file name	Name 18-06-19-02-47-38	Start of a power-on to power-off record, if read as 14:47:38 EDT. Content not decoded.
Main flight record start	2018-06-19 18:50:34	E03	TXT file name	Name [14-50-34]	Start of the main record. Motor-on and take-off not established.
Take off or video start	2018-06-19 18:50:39	E04	MP4 and .info	CaptureDate 14:50:39 EDT	Video start. Take-off is not confirmed from this file.
Landing or video end	2018-06-19 18:54:40 (or 18:54:33.8)	E04	MP4 duration; EndTimeMsec	241 s; 234.833 s	Video end. The two sources differ by about 6 s. Landing not established.
Main flight record event	2018-06-19 18:55:40	E03	TXT stored last write	12:55:40 stored, +6 h	Last write of the main record. Not a landing time.
Power off and correlation	2018-06-19 18:58:24	E05	FLY057 stored last write	12:58:24 stored, +6 h	Stand-in for the end of session 1. Power-off event not decoded.
Second session	2018-06-19 18:59:05 to 19:02:12	E05	FLY058 DAT	Name 02-59-05; last write 13:02:12 stored	Separate record after session 1. No matching TXT or video recovered.

The sources agree in several ways. The file-name clock, the .info header, the container duration and the shared last-write times all point to one session. The video sits inside the main record's span, and that span sits inside FLY057. They conflict or fall short in others: the zip's stored clock is two hours behind the phone clock, the two video durations differ by about six seconds, and extracted file times are not valid UTC. Nothing here is corroborated by the microSD card or an EWF image, because I did not have them, so I could not compare the cached video with the original recording.

7. Conclusion and limitations

The DJI GO 4 data on this phone points to one recording session on 19 June 2018, at about 18:47 to 18:58 UTC, with a roughly four-minute video inside it. I rate that moderate confidence. The limits below are the reasons I go no further.

- **Missing sources.** The raw microSD image, the EWF image and the mobile report were not available to me, so I make no claim about the microSD media, any serial number or any cross-source match.
- **Truncated archive.** The zip is incomplete. Anything stored after Samsung/Music/Over the Horizon.mp3, such as further pictures or a second video cache, was not examined.
- **Records not decoded.** I did not decode the TXT or the DAT files, so the aircraft identity, the route, take-off, landing and GPS positions are not established. A position field exists in the video sidecar, but a single point is not a route.
- **Time-zone uncertainty.** Zip times carry no zone. The EDT zone and the two-hour correction come from one sidecar header and are consistent across five files, but I have not confirmed them from inside the records. The extraction tool also shifted the times on the extracted files.
- **Altered access times.** The access and inode-change times on my working files show 3 November 2026, the day I examined them. They reflect my own extraction and viewing of the working copy and say nothing about 2018. I never opened the original archive for writing.
- **Cache limits.** The cached pictures are DJI-authored app images and the cached video is a lower-quality copy. Neither shows where an aircraft flew or who flew it.

- **Application, device and person.** The artefacts show the DJI GO 4 app on a phone with its clock set to EDT. They do not prove who owned the aircraft, and they do not prove that the person using the phone was flying it.

8. Appendices

Appendix A. Evidence log

ID	Tool and method	Source record, path or image	Observed fact	Screenshot
E01	chmod 444, sha256sum, cp -p, unzip, 7z l, 7z x	Android_Logical.zip (original and working copy)	413,261,824 bytes; SHA-256 49629ec3…d635862 identical for both; archive truncated at Samsung/Music/Over the Horizon.mp3; 373 files extracted	S01, S02, S03, S04
E02	find, ls -l, sha256sum	dji.go.v4 paths: FlightRecord, MCDatFlightRecords, DJI_RECORD, cache/uil-images	TXT, two DAT, MP4 with sidecar files and three cached images with journal located; tutorial assets and the 2017 video excluded	S05, S06, S07, S08, S13
E03	file, xxd, file name and stored time	DJIFlightRecord_2018-06-19_[14-50-34].txt	Binary, 555,776 bytes; name start 14:50:34 EDT = 18:50:34 UTC; content not decoded	S09, S14
E04	exiftool, ffprobe, strings, 7z l -slt	DJI_RECORD/2018_06_19_14_50_39.mp4, .info, analytics/*	1280×720 H.264, 241 s; embedded dates zero; sidecar EDT header; CaptureDate 14:50:39	S10, S11, S12
E05	file name and stored time analysis	FLY057 and FLY058 .DAT	4,002,791 and 1,104,150 bytes; provisional spans 10:46 and 3:07; content not decoded	S07, S12, S14, S15
E06	file, exiftool, cat journal	cache/uil-images/* and journal	Photoshop-authored banner-style images; journal sizes match the files; no GPS tags	S09, S16
E07	not performed	Raw microSD image	Not received	n/a
E08	not performed	EWF image	Not received	n/a

Appendix B. Time-zone check

For the main flight record, 7z l -slt printed Modified = 2018-06-19 12:55:40 under both TZ=UTC and TZ=America/New_York, so the zip stores a plain time with no zone. Under TZ=UTC, ls -l on the extracted file printed 2018-06-19 17:55:40 +0000. Figures S14 and S15 show the stored and extracted times side by side.

```
[Track1]        HandlerDescription              : VideoHandler
[Track1]        GraphicsMode                    : srcCopy
[Track1]        OpColor                         : 0 0 0
[Track1]        CompressorID                    : avc1
[Track1]        SourceImageWidth                : 1280
[Track1]        SourceImageHeight               : 720
[Track1]        XResolution                     : 72
[Track1]        YResolution                     : 72
[Track1]        BitDepth                        : 24
[Track1]        VideoFrameRate                  : 29.236
[ItemList]      Encoder                         : Lavf56.15.102
[Composite]     ImageSize                       : 1280×720
[Composite]     Megapixels                      : 0.922
[Composite]     AvgBitrate                      : 1.92 Mbps
[Composite]     Rotation                        : 0
^C

┌──(kali㉿kali)-[~/case5/working]
└─$ 7z l -slt Android_Logical.zip 2>/dev/null | grep -A5 -E "DJIFlightRecord_2018|FLY05|2018_06_19_14_50_39" | grep -E "Path|Modified"
ls -l --time-style=full-iso $D/DJI/dji.go.v4/FlightRecord/*.txt $D/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/*
Path = Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.info
Modified = 2018-06-19 12:55:42
Path = Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
Modified = 2018-06-19 12:55:42
Path = Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/analytics/2018_06_19_14_50_39.map
Modified = 2018-06-19 12:55:42
Path = Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/analytics/2018_06_19_14_50_39.tlv
Modified = 2018-06-19 12:55:42
Path = Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
Modified = 2018-06-19 12:55:40
Path = Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT
Modified = 2018-06-19 12:58:24
Path = Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-59-05_FLY058.DAT
Modified = 2018-06-19 13:02:12
-rw-rw-r-- 1 kali kali  555776 2018-06-19 13:55:40.000000000 -0400 'android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt'
-rw-rw-r-- 1 kali kali 4002791 2018-06-19 13:58:24.000000000 -0400  android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT
-rw-rw-r-- 1 kali kali 1104150 2018-06-19 14:02:12.000000000 -0400  android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-59-05_FLY058.DAT

┌──(kali㉿kali)-[~/case5/working]
└─$
```

Figure S14. *Stored zip times (7z l -slt) next to extracted file times (ls -l) for the video, the main flight record and the two DAT files.*

```
Path = Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-59-05_FLY058.DAT
Modified = 2018-06-19 13:02:12
-rw-rw-r-- 1 kali kali  555776 2018-06-19 13:55:40.000000000 -0400 'android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt'
-rw-rw-r-- 1 kali kali 4002791 2018-06-19 13:58:24.000000000 -0400  android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT
-rw-rw-r-- 1 kali kali 1104150 2018-06-19 14:02:12.000000000 -0400  android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-59-05_FLY058.DAT

┌──(kali㉿kali)-[~/case5/working]
└─$ cd ~/case5/working; D=android/Android_Logical/sdcard
for z in UTC America/New_York; do echo "== $z"; TZ=$z 7z l -slt Android_Logical.zip 2>/dev/null | grep -A5 "DJIFlightRecord_2018" | grep -E "Path|Modified"
; done
TZ=UTC ls -l --time-style=full-iso "$D/DJI/dji.go.v4/FlightRecord/"*.txt
== UTC
Path = Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
Modified = 2018-06-19 12:55:40
== America/New_York
Path = Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
Modified = 2018-06-19 12:55:40
-rw-rw-r-- 1 kali kali 555776 2018-06-19 17:55:40.000000000 +0000 'android/Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt'

┌──(kali㉿kali)-[~/case5/working]
└─$ cat ../out/04_video_ls.txt ../out/04_video_hashes.txt
strings -n 4 $D/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.info | head -30
android/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD:
total 56460
drwxrwxr-x  3 kali kali     4096 2018-10-03 12:57:36.000000000 -0400 .
drwxrwxr-x 15 kali kali     4096 2018-10-03 12:57:54.000000000 -0400 ..
-rw-rw-r--  1 kali kali     1441 2018-06-19 13:55:42.000000000 -0400 2018_06_19_14_50_39.info
-rw-rw-r--  1 kali kali 57791765 2018-06-19 13:55:42.000000000 -0400 2018_06_19_14_50_39.mp4
drwxrwxr-x  2 kali kali     4096 2018-10-03 12:57:36.000000000 -0400 analytics

android/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/analytics:
total 584
drwxrwxr-x 2 kali kali   4096 2018-10-03 12:57:36.000000000 -0400 .
drwxrwxr-x 3 kali kali   4096 2018-10-03 12:57:36.000000000 -0400 ..
-rw-rw-r-- 1 kali kali 143358 2018-06-19 13:55:42.000000000 -0400 2018_06_19_14_50_39.map
-rw-rw-r-- 1 kali kali 444914 2018-06-19 13:55:42.000000000 -0400 2018_06_19_14_50_39.tlv
2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00  android/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
a42bfeb57c5fff850bc059e10eb9a7b4f7b84887feb71e65464099d1e0fd0b4a  android/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.info
db58496a49d6f8ae82af46338cfcd397f5b7b741b1f726718a69caf5d466dd6a  android/Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/analytics/2018_06_19_14_50_39.map
```

Figure S15. *The same stored time under TZ=UTC and TZ=America/New_York, and the extracted file time under TZ=UTC; the DJI_RECORD listing and hashes follow.*

Appendix C. Screenshot index

Figure	What it shows	Used for
S01	Original read-only, size and SHA-256; unzip failure	Q1, E01
S02	7z l: unexpected end of archive	Q1, E01
S03	Working copy hash match; extraction stops at the mp3	Q1, E01
S04	OneDrive source, 396 MB	Q1, E01
S05, S06	Search for dji.go.v4 paths	Q2, E02
S07	Sizes, times and hashes of TXT, DAT and cache files	Q2, Q5, E02
S08	Tutorial and editor assets excluded	Q2, E02

Figure	What it shows	Used for
S09	file on cached images; TXT header bytes	Q3, Q6, E03
S10	Video, sidecar and analytics hashes; sidecar content (position redacted)	Q4, E04
S11	ExifTool output for the video	Q4, E04
S12	Zip entries for the analytics files	Q4, E04
S13	2017 recording excluded, with hash	Q2, E02
S14, S15	Stored versus extracted times; time-zone check	Q3, Q5, Appendix B
S16	Cache journal and image EXIF	Q6, E06

Commands were run from ~/case5/working on the working copy. The text output of the main commands is saved in the evidence-log folder of the repository named in section 1.